
LECTURE NOTES

Discrete Mathematics notes

Relations

DM549 - Discrete Mathematics

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1 Relations

Definition

Let A, B be sets. A (binary) relation from A to B is a subset of $A \times B$

A relation can be seen as a generalization of a function.

- A *Function* can assign exactly 1 element from B to any element in A
- A *Relation* can assign any element in B to any element in A

1.1 Visualize relations

Let $A = \{1, 2, 3\}$ be a set, and xRy be a relation from A to A .

- Sets

$$R = \{(1, 2), (2, 3), (1, 3)\}$$

- Graph

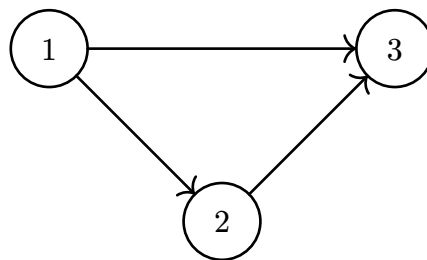


Figure 1: Graph of $R = \{(1, 2), (2, 3), (1, 3)\}$

- Matrix

$$R = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Remember that there exists relations that are neither reflexive or irreflexive, but never both.

Remember that there exists relations that are neither symmetric or anti-symmetric but also both symmetric and anti-symmetric.

1.2 Reflexivity

Definition: Reflexive

A relation R on a set A is called **reflexive** if $(a, a) \in R$ for all $a \in A$.

- Graph $\rightarrow$ self loops on every vertex
- Matrix $\rightarrow$ diagonal is all 1

Definition: Irreflexive

A relation R on a set A is called **irreflexive** if $(a, a) \notin R$ for all $a \in A$.

- Graph $\rightarrow$ no self loops
- Matrix $\rightarrow$ diagonal is all 0

1.3 Symmetry

Definition: Symmetric

A relation R on a set A is called **symmetric** if $(a, b) \in R \implies (b, a) \in R$ for all $a, b \in A$.

- Graph $\rightarrow$ every edge that goes one way also goes the other way (edges are undirected)
- Matrix $\rightarrow$ the matrix equals its own transpose, $R = R^T$

Definition: Anti-symmetric

A relation R on a set A is called **antisymmetric** if $(a, b) \in R \wedge (b, a) \in R \implies a = b$, for all $a, b \in A$.

- Graph $\rightarrow$ no pair of distinct vertices has edges going both ways
- Matrix $\rightarrow$ if $R_{ij} = 1$ and $R_{ji} = 1$ then $i = j$

1.4 Transitivity

Definition (Definition 9.1.5)

A relation R on a set A is called **transitive** if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.

- Graph $\rightarrow$ whenever there is a path $a \rightarrow b \rightarrow c$, there is also a direct edge $a \rightarrow c$

2 Closures

Definition: Closure

Let R be a relation on set A and let P be a property of relations. Then, the *closure* of R w.r.t P is (if it exists) the relation C on A such that

- 1) $R \subset C$,
- 2) C fulfills property P ,
- 3) $C \subset S$ for every S that fulfills 1. and 2, (in place of C)

For some property (like transitive), the closure C is the “minimal” extension“ of R such that R fulfills property P . That being said, if R fulfills P , then $C = R$

Note: anti-symmetric or irreflexive closures does not exist. One can not make an anti-symmetric or irreflexive closure by adding elements.

Definition: Reflexive Closure

The *reflexive closure* of a relation R on a set A is

$$r(R) = R \cup \{(a, a) \mid a \in A\}$$

Definition: Symmetric Closure

The *symmetric closure* of a relation R on a set A is

$$s(R) = R \cup \{(b, a) \mid (a, b) \in R\}$$

Definition: Transitive Closure

The *transitive closure* of a relation R on a set A is

$$t(R) = R^* = \underbrace{\bigcup_{i=1}^{\infty} R^i}_{R \cup R^2 \cup R^3 \cup \dots}$$

in other words When $(a, b), (b, c) \in R$ but $(a, c) \notin R$. Add (a, c) to R until we can no longer find such a, b, c .

3 Combining relations

Definition: Composition of two relations

Let A, B, C be sets, R a relation from A to B and S a relation from B to C .

Then

$$S \circ R = \{(a, c) \mid \exists b : (a, b) \in R \wedge (b, c) \in S\}.$$

If $A = B$, then R^2 denotes $R \circ R$, R^3 denotes $R \circ R \circ R$, etc.

$(a, b) \in R^k$ if and only if one can walk from a to b in k steps along edges (as in a graph)

Example: composition of two relations

Given three sets

$$A = \{1, 2, 3\}, \quad B = \{1, 2, 3, 4\}, \quad C = \{0, 1, 2\}$$

And relations

$$R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}, \quad S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$$

,

Then combination / composition $R \circ S = \{(1, 0), (1, 1), (2, 1), (2, 2), (3, 0), (3, 1)\}$

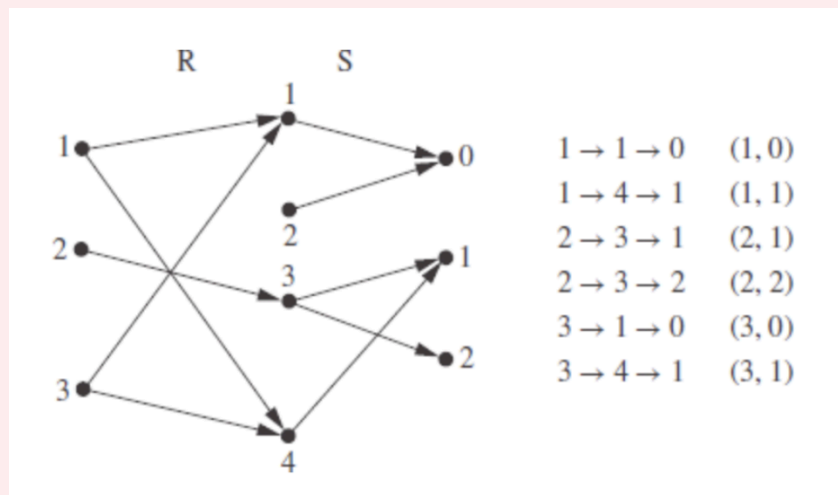


Figure 2: Visual example of composition of two relations

4 Equivalence relation and class

Definition: Equivalence relations

A relation R on a set is called an equivalence relation if it is

- reflexive,
- symmetric, and
- transitive

If this is the case, then for $(a, b) \in R$, a and b are called equivalent.

Definition: Equivalence class

Let R be an equivalence relation on a set A . For $a \in A$.

Then

$$[a]_R = \{b \mid (a, b) \in R\}$$

Is the equivalence class of a w.r.t R .

One can see this as ‘all elements which “equivalent” to a ’

5 Ordering

5.1 Partial order

Definition: Partial order

A relation R on a set A is called a *partial order* if it is

- reflexive,
- anti-symmetric, and
- transitive.

If this is the case, (A, R) is called a partially ordered set (or a poset)

Partial ordering is usually seen as $\leq$ or $\preceq$

5.2 Total order

Definition: Total order

Let $(A, \preceq)$ be a poset, if $a, b \in A$ are comparable, we call $\preceq$ a total order

5.3 Lexicographic order

Definition: Lexicographic order

Let $(A_1, \preceq_1), (A_2, \preceq_2), \dots, (A_n, \preceq_n)$ be partial orders. Then we can define a *lexicographic order*, a partial order, $\preceq$ on $A_1 \times A_2 \times \dots \times A_n$ as follows. For different elements $(a_1, a_2, \dots, a_n), (b_1, b_2, \dots, b_n)$ of $A_1 \times A_2 \times \dots \times A_n$ that are not equal, $(a_1, a_2, \dots, a_n) \preceq (b_1, b_2, \dots, b_n)$ holds if and only if

- $a_1 \prec b_1$, or
- there exists an $i > 0$ such that $a_1 = b_1, a_2 = b_2, \dots, a_i = b_i$ and $a_{i+1} \preceq_{i+1} b_{i+1}$.

Example: Lexicographic order

Words (sets of letters) are lexicographically ordered in most dictionaries.

hate $\preceq$ hope $\preceq$ love